Homework 1 - Questions and worked answers

Mathematical Foundations: powers, logarithms, ratios and quadratic equations

Each original problem is followed by its worked solution. The source notebook labels two problems 1 and skips 6; these labels are retained. The sound experiment requires students' own measurements. The numerical solution uses only the notebook's illustrative dataset.

1. Ribosomes and protein production during cell growth

Original problem

An *E. coli* cell contains approximately $3 \cdot 10^6$ proteins and $2.5 \cdot 10^4$ ribosomes. Assume that an average protein consists of **300 amino acids** and that the cell doubles in **50 minutes**.

Questions

a) How many new proteins must the cell produce per second in order to double its protein content in 50 minutes?

b) Assuming that all ribosomes contribute equally, how many proteins must one ribosome produce per second?

c) From this result, estimate how many amino acids a ribosome must add to a growing protein chain per second.

d) Compare your result with a typical experimentally measured translation rate of about 15–20 amino acids per second in *E. coli*. Is your estimate reasonable?

e) Think: If your estimate differs from the experimental value, what assumptions in our simple calculation might explain the difference?

> Keep the numbers in scientific notation for as long as possible. The aim is to reason with powers of ten and ratios rather than repeatedly entering numbers into a calculator.

Worked solution

The doubling time is $T = 50 \times 60 = 3000\,\text{s}$. The cell must make an additional 3×10^6 proteins during this time.

a) Average protein production rate for the cell

$$\frac{3 \times 10^6}{3 \times 10^3} = 10^3 \text{ proteins/s.}$$

b) Average production rate per ribosome

$$\frac{10^3}{2.5 \times 10^4} = 0.04 \text{ proteins per ribosome per second.}$$

This corresponds to one protein every 25 s per ribosome.

c) Amino-acid addition rate

$$0.04 \times 300 = 12 \text{ amino acids per ribosome per second.}$$

d) This is a reasonable order-of-magnitude estimate: 12 is somewhat below the stated experimental range of 15–20 amino acids/s, but on the same biological scale.

e) The protein count, mean protein length and doubling time are approximate. Not every ribosome is actively translating; initiation and termination also take time. Protein degradation requires additional synthesis. Moreover, ribosome numbers change as the cell grows, whereas this estimate divides a time-averaged output by one fixed ribosome count. These effects need not all change the estimate in the same direction. For example, if only 60–80% of the assumed ribosomes were active, the required rate per active ribosome would be $12/0.8 = 15$ to $12/0.6 = 20$ amino acids/s, with the other assumptions fixed.

1. Powers and orders of magnitude

Original problem

A bacterium has a typical length of about

$$2 \times 10^{-6} \text{ m},$$

while a human has a height of roughly

$$2 \text{ m}.$$

Questions

a) Without using a calculator, estimate how many times larger a human is than the bacterium.

b) A virus has a typical diameter of about

$$10^{-7} \text{ m}.$$

Approximately how many virus particles placed next to each other would span the length of the bacterium?

c) Suppose another biological structure is 1000 times smaller than the bacterium. Write its approximate size in metres using scientific notation.

d) Explain briefly why powers of ten are useful when comparing biological length scales.

> **Hint:** When dividing powers with the same base, subtract the exponents.

Worked solution

a) Human height divided by bacterial length:

$$\frac{2\,\text{m}}{2 \times 10^{-6}\,\text{m}} = 10^{6}.$$

A human is approximately **one million times longer**.

b) $\dfrac{2 \times 10^{-6}}{10^{-7}} = 2 \times 10^{1} = 20$ virus particles span the bacterial length.

c) $\dfrac{2 \times 10^{-6}\,\text{m}}{10^{3}} = 2 \times 10^{-9}\,\text{m} = 2\,\text{nm}.$

d) Powers of ten express very different sizes compactly. Ratios are easy to calculate by subtracting exponents, making differences in scale immediately visible.

2. Exponents and logarithms

Original problem

A population of bacteria doubles every generation. Starting with one bacterium, after n generations the population is

$$N = 2^n.$$

Questions

a) How many bacteria are present after 3, 4 and 5 generations? Do this without a calculator.

b) You observe approximately 1000 bacteria. Estimate how many generations must have occurred.

Try first to answer this by knowing or estimating nearby powers of 2.

c) We can solve

$$N = 2^n$$

for n using logarithms.

Take the logarithm of both sides and derive

$$n = \frac{\ln N}{\ln 2}.$$

d) Would you obtain a different value for n if you used log instead of ln? Explain.

> The important idea is that a logarithm answers the question: **To what power must a base be raised to obtain a given number?**

Worked solution

a) With $N = 2^n$:

Generations n	Bacteria N
3	$2^3 = 8$
4	$2^4 = 16$
5	$2^5 = 32$

b) Since $2^{10} = 1024$, approximately **10 generations** produce about 1000 bacteria.

c) Take the logarithm of both sides:

$$\ln N = \ln(2^n) = n \ln 2 \quad \Longrightarrow \quad \boxed{n = \frac{\ln N}{\ln 2}}.$$

For $N = 1000$, this continuous interpolation gives $n \approx 9.966$. In the model of synchronized, discrete divisions, populations occur at integer generations; the first integer generation reaching at least 1000 is 10.

d) No. Using the same logarithm base in numerator and denominator gives the same answer:

$$\frac{\log_{10} N}{\log_{10} 2} = \frac{\ln N / \ln 10}{\ln 2 / \ln 10} = \frac{\ln N}{\ln 2}.$$

3. Exponential decay and the natural logarithm

Original problem

Many processes in biology can be described approximately by

$$C(t) = C_0 e^{-kt},$$

where C_0 is the initial concentration and $k > 0$.

Questions

a) What is $C(0)$? Explain your answer without calculating numerically.

b) At the half-life $t_{1/2}$,

$$C(t_{1/2}) = \frac{C_0}{2}.$$

Starting from

$$\frac{C_0}{2} = C_0 e^{-kt_{1/2}},$$

derive an expression for $t_{1/2}$.

c) Why does $\ln 2$ appear in the result?

d) Suppose k is doubled. What happens to the half-life?

Do not insert numbers; reason from your formula.

Worked solution

The model is $C(t) = C_0 e^{-kt}$, with $k > 0$.

a) At $t = 0$, the exponent is zero and $e^0 = 1$, so

$$\boxed{C(0) = C_0}.$$

b) For $C_0 > 0$, divide the half-life equation by C_0, then take logarithms:

$$\frac{1}{2} = e^{-kt_{1/2}}, \qquad \ln\left(\frac{1}{2}\right) = -kt_{1/2}.$$

Since $\ln(1/2) = -\ln 2$,

$$\boxed{t_{1/2} = \frac{\ln 2}{k}}.$$

c) Halving introduces the factor $1/2$. The natural logarithm undoes the exponential, and $\ln(1/2) = -\ln 2$. This is why $\ln 2$ appears.

d) Doubling k halves the half-life:

$$t'_{1/2} = \frac{\ln 2}{2k} = \frac{t_{1/2}}{2}.$$

As a unit check, kt must be dimensionless, so k has units of inverse time and $\ln 2/k$ has units of time.

4. Ratios and scaling

Original problem

Consider two spherical cells. Cell B has **twice the radius** of cell A:

$$R_B = 2R_A.$$

For a sphere,

$$A = 4\pi R^2$$

and

$$V = \frac{4}{3}\pi R^3.$$

Questions

a) Determine the ratio

$$\frac{A_B}{A_A}.$$

Do **not** calculate the individual areas.

b) Determine

$$\frac{V_B}{V_A}.$$

Again, do not calculate the individual volumes.

c) Compare the surface-to-volume ratios

$$\frac{A}{V}$$

of the two cells. Which cell has more surface area available per unit volume?

d) Give one reason why surface-to-volume ratio can matter in biology.

Try to cancel common factors before doing any arithmetic.

Worked solution

Given $R_B = 2R_A$, $A = 4\pi R^2$ and $V = \frac{4}{3}\pi R^3$:

a) Surface areas

$$\frac{A_B}{A_A} = \frac{4\pi(2R_A)^2}{4\pi R_A^2} = \boxed{4}.$$

b) Volumes

$$\frac{V_B}{V_A} = \frac{\frac{4}{3}\pi(2R_A)^3}{\frac{4}{3}\pi R_A^3} = \boxed{8}.$$

c) Surface-to-volume ratios

$$\frac{A}{V} = \frac{4\pi R^2}{\frac{4}{3}\pi R^3} = \frac{3}{R}.$$

Therefore,

$$\frac{(A/V)_B}{(A/V)_A} = \frac{R_A}{R_B} = \frac{1}{2}.$$

Cell A has twice as much surface area per unit volume as cell B. Cell B has more total surface area, but its volume increases by a larger factor.

d) Nutrients and waste can cross the cell membrane, while metabolic demand is associated with the cell interior. A smaller surface-to-volume ratio can limit the exchange capacity available per unit cell volume, if other factors are comparable.

5. Quadratic equations and physical solutions

Original problem

The height of an object can be described in convenient units by

$$h(t) = 1 + 4t - t^2.$$

The object reaches the ground when $h = 0$.

Questions

a) Write down the quadratic equation that determines the time at which the object reaches the ground.

b) Solve the equation. You may complete the square, factor if possible, or use the quadratic formula

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

c) You obtain two mathematical solutions. Which solution describes the physical event after the experiment starts at $t = 0$?

d) What does the other solution tell you?

Explain why **solving an equation** and **interpreting its solutions** are two separate steps.

Worked solution

The given height is $h(t) = 1 + 4t - t^2$ in the problem's convenient units.

a) Ground contact requires

$$1 + 4t - t^2 = 0, \qquad \text{or} \qquad t^2 - 4t - 1 = 0.$$

b) Complete the square:

$$t^2 - 4t = 1, \qquad (t-2)^2 = 5,$$

$$\boxed{t = 2 \pm \sqrt{5}}.$$

The roots are approximately 4.236 and -0.236, in the specified time units. Substitution verifies both: $1 + 4(2 \pm \sqrt{5}) - (2 \pm \sqrt{5})^2 = 0$.

c) The ground contact after the experiment starts is $\boxed{t = 2 + \sqrt{5}}$, because it is positive.

d) The negative root describes a crossing before $t = 0$ in the backward extension of the model, not an event after the experiment starts. Algebra gives the roots; the physical time interval selects the relevant one.

7. Home experiment - sound level versus distance

Original problem

For this experiment, use the **phyphox** app on your smartphone and measure the sound level in **decibels (dB)**.

Use a second phone, computer or loudspeaker to produce a continuous tone. Keep its position and volume unchanged throughout the experiment.

1. Check the background level

First measure the sound level in the room with the loudspeaker switched **off**.

A quiet room might, for example, give a value around 40 dB. Record your own value:

$$L_{\text{background}} = \qquad \text{dB}.$$

If phyphox or your phone requires a calibration offset, you may adjust the displayed value so that the quiet-room reading corresponds approximately to a plausible room level such as 40 dB.

Important: Such a calibration changes the **offset** of all dB values but not how rapidly the measured level changes with distance. Therefore the slope determined below is unaffected by a constant calibration offset.

Also, do **not** simply subtract 40 dB from another dB reading. Decibels are logarithmic quantities.

2. Measure at different distances

Place the measuring phone at approximately

$$r, \qquad 2r, \qquad 3r, \qquad 4r$$

from the loudspeaker.

Choose r so that the loudspeaker is clearly louder than the background at all four positions.

Record the actual distance and the measured sound level:

distance r (m)	sound level L (dB)

3. What should we expect?

For an ideal point source in free space, sound intensity follows approximately

$$I \propto \frac{1}{r^2}.$$

The sound level in decibels is defined from an intensity ratio by

$$L = 10 \log_{10} \left(\frac{I}{I_0} \right).$$

Insert

$$I = \frac{C}{r^2}$$

and show that the distance dependence can be written as

$$L = L_0 - 20 \log_{10}(r),$$

where L_0 contains all terms that do not depend on distance.

Questions

a) Why do we no longer need to take the logarithm of the measured sound level? What has the dB scale already done for us?

b) If you plot L in dB against $\log_{10}(r)$, what shape should you expect?

c) What slope do you expect for an ideal point source?

d) What happens to the sound level when the distance is doubled? Use the equation above to predict the change before looking at your measurements.

e) Why might your measurements in a room differ from the ideal prediction? Give at least two reasons.

> The aim is not to reproduce the ideal law perfectly. Reflections, the directional properties of the loudspeaker and microphone, background noise, and the finite size of the source all affect the measurement.

Worked solution

Expected behaviour, not a prescribed experimental result. Students should report their own background level, distances and sound levels. A quiet-room reading is not fixed at 40 dB; assigning a plausible value is an offset adjustment, not a traceable calibration.

Deriving the distance law

Use a reference distance r_0 so that logarithms have dimensionless arguments. For an ideal point source,

$$I(r) = I(r_0)\left(\frac{r_0}{r}\right)^2.$$

Insert this into the definition of sound level:

$$\begin{aligned} L(r) &= 10\log_{10}\left[\frac{I(r_0)}{I_0}\left(\frac{r_0}{r}\right)^2\right] \\ &= 10\log_{10}\left(\frac{I(r_0)}{I_0}\right) - 20\log_{10}\left(\frac{r}{r_0}\right). \end{aligned}$$

Thus

$$\boxed{L(r) = L(r_0) - 20\log_{10}(r/r_0)}.$$

Taking $r_0 = 1\,\mathrm{m}$ yields the notebook's expression when r is inserted as its numerical value in metres. Its constant L_0 is the reference-distance level; I_0 is the reference intensity in the decibel definition.

a) The dB scale has already taken a logarithm of an intensity ratio. We plot the measured level L directly, rather than taking another logarithm of it.

b) Plotting L against $x = \log_{10}(r/r_0)$ gives a **straight line**.

c) Its ideal slope is $-20\,\mathrm{dB}$ **per decade of distance**. Increasing distance by a factor of 10 reduces the level by 20 dB.

d) Doubling distance gives

$$\Delta L = L(2r) - L(r) = -20\log_{10} 2 \approx \boxed{-6.02\,\mathrm{dB}}.$$

The intensity decreases to one quarter. The expected changes relative to the first measurement are:

Distance	Relative intensity	Level change
r	1	0 dB
$2r$	1/4	−6.02 dB
$3r$	1/9	−9.54 dB
$4r$	1/16	−12.04 dB

e) Possible deviations include wall reflections and standing waves, background sound, loudspeaker and microphone directionality, finite source size and near-field effects, distance uncertainty, and automatic microphone gain changes. Two well-explained reasons are sufficient. In a room, a phone measures sound pressure; relating its level to the ideal intensity law assumes a suitable freely propagating sound field.

8. Programming - test the distance law

Original problem

Enter the **actual distances** and the corresponding sound levels measured by phyphox.

The measured quantity is already logarithmic because it is expressed in dB. Therefore, instead of plotting $\ln I$ against $\ln r$, we plot

$$L$$

against

$$\log_{10}(r).$$

For an ideal point source we expect

$$L = L_0 - 20\log_{10}(r).$$

```
import numpy as np
import matplotlib.pyplot as plt

# Replace these with YOUR measured distances in metres
r = np.array([0.25, 0.50, 0.75, 1.00])

# Replace these with YOUR phyphox sound levels in dB
L = np.array([72.0, 66.0, 62.5, 60.0])

plt.figure(figsize=(7, 4))
plt.plot(r, L, "o-")
plt.xlabel("distance r (m)")
plt.ylabel("sound level L (dB)")
plt.title("Measured sound level")
plt.grid()
plt.show()
```

Now transform **only the distance axis**:

$$x = \log_{10}(r).$$

If the inverse-square law is valid, L plotted against x should approximately form a straight line with slope

$$-20 \text{ dB/decade}.$$

```
x = np.log10(r)

slope, intercept = np.polyfit(x, L, 1)
fit = intercept + slope*x

plt.figure(figsize=(7, 4))
plt.plot(x, L, "o", label="measurement")
plt.plot(x, fit, label=f"fit: slope = {slope:.1f} dB/decade")
plt.xlabel("log10(r / m)")
plt.ylabel("sound level L (dB)")
plt.title("Sound level versus logarithmic distance")
plt.grid()
plt.legend()
plt.show()

print("Measured slope =", slope, "dB/decade")
print("Ideal point-source slope = -20 dB/decade")
```

Questions

a) Is the graph approximately a straight line?

b) What slope do you obtain? Compare it with the ideal value of

$$-20 \text{ dB/decade}.$$

c) From your fitted slope, predict how many dB the sound level changes when the distance is doubled. Compare this with the ideal prediction of approximately **-6 dB**.

d) Suppose you add a constant calibration offset of +5 dB to every measurement. What happens to the graph? What happens to its slope?

e) Why is it incorrect simply to subtract the background value in dB from the measured value in dB?

Optional: If the loudspeaker signal becomes comparable with the room background, the background can be removed on a linear intensity scale. If L_{total} is measured with the loudspeaker on and $L_{\text{background}}$ with it off, an estimate of the source contribution is

$$L_{\text{source}} = 10 \log_{10} \left(10^{L_{\text{total}}/10} - 10^{L_{\text{background}}/10} \right).$$

You do not need this correction if the loudspeaker is clearly louder than the background.

Worked solution

Use the actual measurements for a student's answer. The following is a **worked example using the data already supplied in the notebook**, not a new measurement.

Distance r (m)	$x = \log_{10}(r/1\,\text{m})$	Sound level L (dB)
0.25	−0.60206	72.0
0.50	−0.30103	66.0
0.75	−0.12494	62.5
1.00	0	60.0

a) For these example data, L plotted against logarithmic distance is almost a straight line.

b) The least-squares fit $L = b + sx$ gives

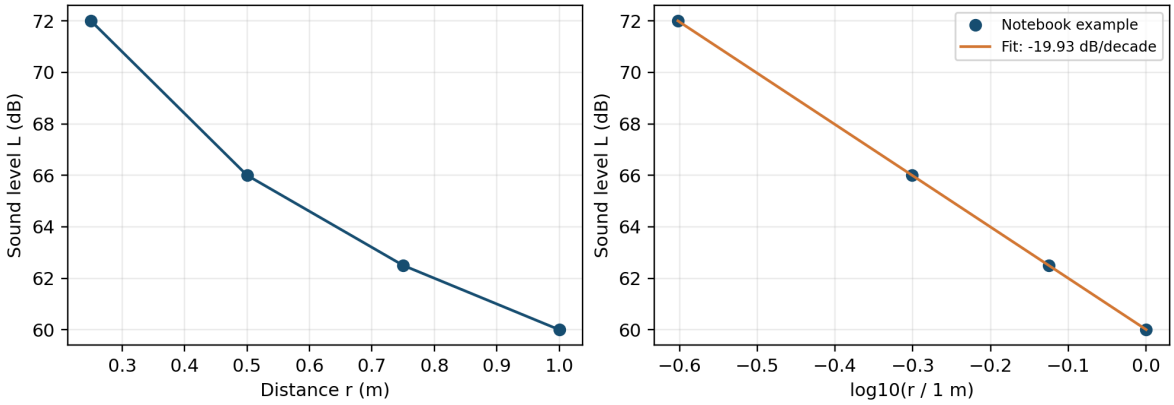
$$\boxed{s \approx -19.925\,\text{dB/decade}}, \qquad \boxed{b \approx 60.004\,\text{dB}}.$$

Rounded sensibly, the slope is $-19.9\,\text{dB/decade}$, very close to the ideal -20. Such close agreement is a property of this illustrative dataset; real room measurements may differ appreciably.

c) A distance doubling changes x by $\log_{10} 2$, so

$$\Delta L = s \log_{10} 2 \approx -5.998\,\text{dB} \approx -6.00\,\text{dB}.$$

This agrees closely with the ideal prediction of $-6.02\,\text{dB}$.



d) Adding $+5\,\text{dB}$ shifts every point vertically upward by 5 dB. The intercept becomes approximately $65.004\,\text{dB}$, while the slope and the predicted change on doubling distance remain unchanged.

Background correction and example code

e) Subtracting dB readings forms a logarithm of an intensity **ratio**, not an intensity difference:

$$L_{\text{total}} - L_{\text{background}} = 10 \log_{10} \left(\frac{I_{\text{total}}}{I_{\text{background}}} \right).$$

To isolate the source, subtract on a linear scale. Assuming source and background contributions add in mean-square pressure (or the corresponding intensity model),

$$\frac{I_{\text{source}}}{I_0} = 10^{L_{\text{total}}/10} - 10^{L_{\text{background}}/10},$$

and therefore

$$\boxed{L_{\text{source}} = 10 \log_{10} \left(10^{L_{\text{total}}/10} - 10^{L_{\text{background}}/10} \right)}.$$

For example, $L_{\text{total}} = 60\,\text{dB}$ and $L_{\text{background}} = 40\,\text{dB}$ give $L_{\text{source}} \approx 59.96\,\text{dB}$, **not 20 dB**. This subtraction requires $L_{\text{total}} > L_{\text{background}}$ and becomes unreliable when the levels are very close. A constant calibration offset still changes only the resulting level offset.

Reproducible analysis

```
import numpy as np
import matplotlib.pyplot as plt

# Notebook example; replace with your own measurements.
r = np.array([0.25, 0.50, 0.75, 1.00])  # metres
L = np.array([72.0, 66.0, 62.5, 60.0])  # dB
r0 = 1.0                                # metre

x = np.log10(r / r0)
slope, intercept = np.polyfit(x, L, 1)
delta_L = slope * np.log10(2)

print(f"Slope: {slope:.3f} dB/decade")
print(f"Intercept: {intercept:.3f} dB")
print(f"Change on doubling: {delta_L:.3f} dB")

plt.plot(x, L, "o", label="Example data")
plt.plot(x, intercept + slope*x, label="Linear fit")
plt.xlabel("log10(r / r0)")
plt.ylabel("Sound level L (dB)")
plt.grid(True)
plt.legend()
plt.show()
```

Expected output: slope $-19.925\,\mathrm{dB/decade}$, intercept $60.004\,\mathrm{dB}$, doubling change $-5.998\,\mathrm{dB}$.